B.Tech Project – Weekly Progress Record

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Date	Work Done	Student Signature	Supervisor signature
07/08/2026	Project allocation and literature survey on urban cyclist safety. Studied SimRA crowdsourced dataset structure and German StVO § 5(4) 1.5 m statutory passing clearance regulations.		
14/08/2026	Reviewed existing bicycle proximity detection systems (ProxiCycle, OpenBikeSensor). Identified core research gaps: lack of roll tilt compensation and uniform severity treatment.		
21/08/2026	Formulated system functional and non-functional requirements. Investigated microscopic traffic simulation tools, selecting Eclipse SUMO for sublane lateral dynamics modeling.		
28/08/2026	Designed software system architecture: Go REST API backend, PostgreSQL database schema for ride persistence, Redis caching layer, and Leaflet.js interactive canvas map interface.		
04/09/2026	Preprocessed and filtered empirical bicycle ride traces from the open SimRA dataset. Analyzed GPS trajectories and accelerometer deceleration profiles for near-miss indicators.		
11/09/2026	Implemented web dashboard interface for ingesting bicycle telemetry logs, rendering spatial paths on Leaflet maps, and plotting speed and longitudinal acceleration time-series.		
18/09/2026	Integrated Python TraCI client with Eclipse SUMO to orchestrate dynamic vehicle movements. Validated initial single-corridor simulation loop and lateral clearance calculations.		
25/09/2026	Configured multi-modal road network. Modeled four distinct vehicle interaction classes (bicycle-bicycle, bicycle-car, bicycle-truck/bus, and bicycle-motorbike).		
01/10/2026	TraCI simulation runner review. Formulated multi-cycle traffic generation plan driven by real-world counts and researched domestic Indian component sourcing for hardware.		
09/10/2026	Constructed and calibrated 10 km Berlin arterial digital twin (Ernst-Reuter-Platz to Oberbaumbrücke) using official Berlin Senate 12-hour automated traffic census counts.		
16/10/2026	Trained XGBoost 3-class conflict classifier on SimRA kinematic features (90.6% F1-score vs 74.4% static rule). Verified domestic INR 3,494 hardware BOM and power regulation design.		
23/10/2026	Conducted sublane lateral resolution sensitivity ablation (0.2 m vs 0.5 m). Evaluated 1,048 simulated encounters. Added dynamic multi-bicycle tracking with CSV telemetry export.		
06/11/2026	Compiled BTP Part 1 comprehensive thesis report, circuit schematics, tilt-compensation mathematical proof ($d_{true} = d_{measured} \cdot \cos\varphi$), and Dhanbad domestic component BOM.		
12/11/2026	Finalized thesis report and verified Turnitin similarity index at 8% (well within 20% limit). Formulated presentation slides and progress documentation for project evaluation.		